Client Server Model.

① Gaming server ② Mail server.

Advantages

i) Central resource management
ii) Scalability
iii) Flexibility of Reach

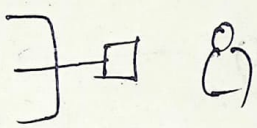
Disadvantage

i) Single point of failure
ii) Network congestion
iii) Security risk

Load Balancer

i) Balancing Lot of customers

- Load Balancer
  - physical hardware
  - software
  - just a snippet



Types of Balancing

1) Least connection 2) Round Robin scheduling
3) weight Round Robin 4) IP hashing

Advantages

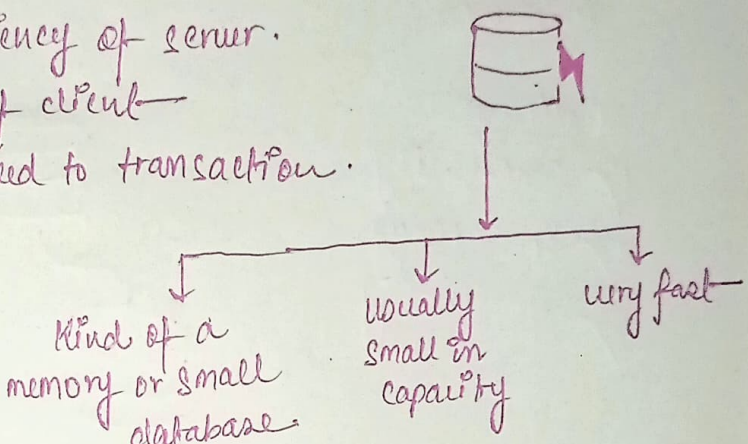
1) Traffic is distributed and processed in parallel
2) Your design becomes easy to scale up.
3) High availability

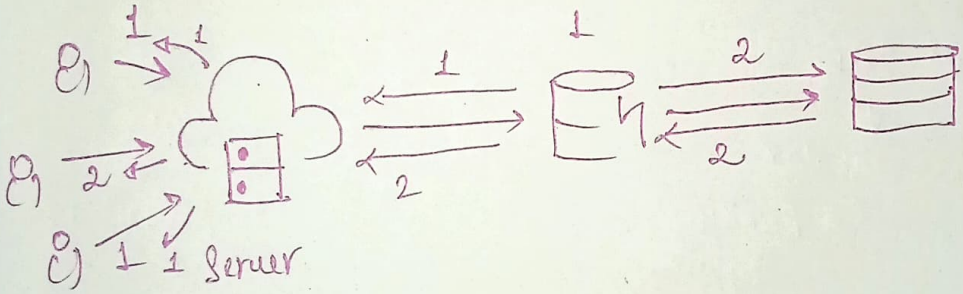
Disadvantage

1) A load balancer can get very complex and not easy to setup.
2) The load balancer itself becomes a single point of failure.
3) If load balancer is compromised, system can go down.

Caching

1) Increase Efficiency of server.
2) Saves time of client
3) Increased speed to transaction.





Important caching terms.

① Cache Hit : you are able to find recource in your cache.

② Cache Miss : unable to find resource in your cache.

↑↑ value of cache Hit, better is the system.

Cache Eviction methods.

① Random eviction ② FIFO : First In First Out

③ LFU : Least Frequently used. ④ LRU : Least Recently used.

Things to be careful about

1) consistency of resources.

2) Concurrence: caches in different regions could get out of sync.

3) Security: session and credential should be cached with caution.

Database

1) Efficient data retrival

2) Data integrity & consistency

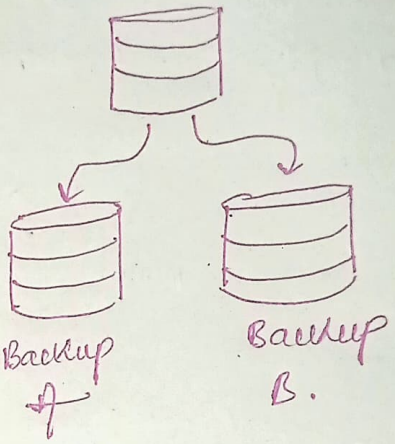
3) Handle complex query

Challenge

1) Scalability

2) consistency

3) Availability



NO-SQL vs SQL

NO-SQL → Not a tabular one → insta comment

SQL → Tabular → Banking system.